

The Positive Markoff–Vieta Tree: Unique Descent, Global Termination, and the Arithmetic Route-A Boundary

Route-A structural certificate C193

Abstract

We orient the entire positive Markoff graph, modulo coordinate permutations, by replacing the unique largest coordinate with the other Vieta root. Every non-root normalized solution has a positive parent of strictly smaller height; integer descent terminates at $(1, 1, 1)$, reverse edges generate every positive solution, and the quotient graph is a rooted tree. A producer-independent bounded scan and separate symbolic reconstruction test the exact convention without replacing the all-solution proof. The integer cubic gives intrinsic Diophantine structure but no rational-prime primitive carrier, prime-power repetition or logarithmic clock. Strict descent also removes every non-root periodic orbit, so the candidate remains below Route A despite its arithmetic origin.

Keywords: Markoff equation; Vieta involution; infinite descent; Diophantine dynamics; rooted tree; Route-A obstruction.

中文摘要

本文在坐标置换商上，为全部正整数Markoff三元组定义最大坐标Vieta下降。除根 $(1, 1, 1)$ 外，每个规范化解都有唯一最大坐标；替换为二次方程的另一根得到严格更低的正整数父节点。因此下降有限终止，反向边生成全部正解，商图构成唯一父节点的有根树。精确整数扫描与独立符号复核只作约定哨兵。该系统具有内生丢番图结构，却没有有理素数本原载体、素数幂重复或对数时钟；严格下降也排除了非根周期轨道。

关键词：Markoff方程；Vieta对合；无穷递降；有根树；算术动力学。

1 Frozen Diophantine dynamics

Let

$$P(x, y, z) = x^2 + y^2 + z^2 - 3xyz.$$

A normalized Markoff triple is a positive integral zero with $x \leq y \leq z$. For example $R_z(x, y, z) = (x, y, 3xy - z)$. Direct expansion gives $P \circ R_z = P$ and $R_z^2 = \text{id}$; the other maps follow by symmetry. As a quadratic in z , the two roots z, z' obey

$$z + z' = 3xy, \quad zz' = x^2 + y^2, \quad z' = 3xy - z. \quad (1)$$

The classical equation and reduction genealogy go back to Markoff [1]; Bombieri gives a modern tree account [2].

2 Unique parent, termination, and completeness

First, every non-root normalized solution has a unique maximum. If $y = z$, the equation becomes

$$(3x - 2)y^2 = x^2.$$

For $x \geq 2$ the quotient $x^2/(3x - 2)$ is smaller than x , contradicting $y^2 \geq x^2$; for $x = 1$ it forces $y = 1$, the root.

Now let $q(t) = t^2 - 3xyt + x^2 + y^2$. At $t = y$,

$$q(y) = x^2 - (3x - 2)y^2 \leq 0.$$

Together with (1), this places y between the two positive roots. Hence every non-root solution satisfies

$$0 < z' = 3xy - z \leq y < z. \quad (2)$$

Define $D(x, y, z) = \text{sort}(x, y, z')$ and fix $(1, 1, 1)$. The integer height $h = z$ strictly decreases, so every orbit terminates. The only terminal point is the root, since (2) supplies a parent otherwise.

No Vieta edge is lost by this orientation. Replacing either nonmaximal coordinate gives

$$(3yz - x) - z = z(3y - 1) - x > 0, \quad (3xz - y) - z = z(3x - 1) - y > 0.$$

Thus both nonparent mutations ascend strictly, while the largest-coordinate mutation is the unique descending edge.

The source-locked Vieta-orbit statement records that all positive triples are obtained from the root by coordinate permutations and Vieta involutions [3]. Reversing D therefore generates every normalized positive solution. The descending coordinate is unique, and the preceding inequalities exhaust all other Vieta edges, so each non-root vertex has exactly one parent. Connectivity and strict height exclude cycles: the normalized solution graph is a rooted tree. Repeated coordinates merely collapse duplicate outgoing mutations at the first two levels $(1, 1, 1) \rightarrow (1, 1, 2) \rightarrow (1, 2, 5)$.

3 Evidence and the exact Route-A stop

The finite certificate contains 513 tree rows through depth ten, all 15 solutions with largest coordinate at most 2,000, and 19 complete descent traces. Depth-ten rows retain their one-step depth-eleven frontier children, so the census is not misdescribed as a closed finite tree. A producer-independent checker solves a different quadratic variable and passes 8,417 assertions; a separate SymPy path passes 8,418 checks; replay is byte exact; 156 repaired-hash and one stale-hash attacks are rejected. The census is regression, not proof of global completeness.

The integer cubic and Vieta maps are intrinsic Diophantine data, so A0 is weak rather than absent. They do not select rational primes as primitive orbits, prime powers as repetitions or $\log p$ as time. Moreover height strictly decreases, so the autonomous descent has no non-root periodic orbit. A tree adjacency operator is only a formal choice; no target divisor, functional equation or source-native target quantization follows. Thus

$$(A0, A1, A2, A3, A4) = (\text{WEAK ARITHMETIC}, \text{FAIL}, \text{FAIL}, \text{FAIL}, \text{FORMAL HINT}),$$

overall ROUTE_A_REJECTED, Route B false, under NO_BAD_EULER_OR_ROOT_NUMBER.

Open-problem and scope firewall. Unique parenthood of a triple does not say that its largest Markoff number determines the other coordinates. The Frobenius uniqueness conjecture remains open and is not used. The modular strong-approximation program of Ref. [3] supplies context only: this package uses no reduction modulo a prime, local arithmetic datum, Euler factor, root number or automorphy claim. It claims neither classical priority nor external review.

Revision-round focus. Round 2 adds source ownership, independent evidence, the Frobenius/modular firewall and strict Route-A grading.

Declarations

Data and code availability. Package-local exact evidence and deterministic code accompany this manuscript.

Ethics. No human, animal, clinical, personal or sensitive data are used; approval is not applicable.

Author contributions (CRediT). This anonymous certificate records Conceptualization, Formal analysis, Software, Validation, Writing—original draft, and Writing—review and editing. AI systems are not authors.

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AI-use disclosure. An AI coding assistant supported drafting and exact-code development; it was not an external reviewer or independent error process.

References

- [1] A. Markoff, “Sur les formes quadratiques binaires indéfinies,” *Math. Ann.* 15 (1879), 381–406. DOI: 10.1007/BF02086269.
- [2] E. Bombieri, “Continued fractions and the Markoff tree,” *Expo. Math.* 25(3) (2007), 187–213. DOI: 10.1016/j.ex-math.2006.10.002.
- [3] J. Bourgain, A. Gamburd and P. Sarnak, “Markoff triples and strong approximation,” *C. R. Math.* 354(2) (2016), 131–135. DOI: 10.1016/j.crma.2015.12.006.